

CSE 4108 - Structured Programming I Lab

Lab 4

OBJECTIVES:

- i. Exploring Conditional statements
- ii. Introduction to logical operators
- iii. Introduction to vjudge

Task 1:

Write a program that takes two numbers as input and produces the output according to the description below:

- It prints "The numbers are equal" if both numbers are same
- It prints "First number is larger than second one" if first number is greater than second number
- It prints "First number is smaller than second one" if first number is smaller than second number

Task 2:

Write a program that reads three numbers from the users and prints the largest number among them.

Try to minimize the number of comparison needed to solve the problem.

[**It is possible to solve this problem using 3 comparisons only. If you are allowed to use extra variable, then it can be solved using 2 comparisons!**]

Optional Task: Try to find the smallest number among them.

Sample execution is provided below:

```
Enter 3 numbers: 12 25 3
25 is the largest number
Process returned 0 (0x0)   execution time : 6.047 s
Press any key to continue.
```

Figure 1: Sample execution

Task 3:

Assume you are writing a program to calculate the admission test result of Imperial University of Technology (IUT). The admission test contains 100 questions and each question provides 1 mark for correct answer. If anyone provides a wrong answer, 0.25 marks are deducted from his score.

You will be provided with two information:

- The number of total questions answered by a student
- The number of wrong answer

Your goal is to calculate the final score of the student and print his admission status based on the condition provided below:

- The student will get a self financed admission, if his score is more than 40 but less than 70
- The student will get partial Scholarship, if his final score is more or equal to 70 but less than 95
- The student will get the Full Scholarship, if his final score is 95 or above
- The student will not get the admission, if his score is 40 or below

Sample execution is provided below:

```
Enter the number of questions answered by the student: 10
Enter the number of wrong answers: 4

The final score of the student is: 5
The student is not selected for admission.
Process returned 0 (0x0)    execution time : 4.281 s
Press any key to continue.
```

Figure 2: Example 1

```
Enter the number of questions answered by the student: 100
Enter the number of wrong answers: 4

The final score of the student is: 95
The student is selected for full scholarship.
Process returned 0 (0x0)    execution time : 2.734 s
Press any key to continue.
```

Figure 3: Example 2

Task 4:

Introduction to vjudge:

- Create an account or login at **<https://vjudge.net>**
 - In your profile page, set the nickname as your student ID
 - Participate in the following contest using the information given below:
Link: **<https://vjudge.net/contest/578308>**
Password: lab41a
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